

Deep-AMR Face-Centered $\nabla \cdot B$ Bug Fix Guide

AthenaK Gotham / MHD AMR validation notes

April 28, 2026

1 Purpose

This guide documents the face-centered MHD AMR bug that produced nonzero $\nabla \cdot \mathbf{B}$ in deep moving-AMR tests, what the code did before PR 732, what the current code does, and how the current fix was validated. The important point is that the fix is not a divergence cleaner or projection. It preserves the constrained-transport AMR construction by making sure that the face fields used by Toth–Roe prolongation and restriction are the correct face fields, owned by the correct AMR path, at the correct time in the mesh rebuild.

2 Discrete Constraint

AthenaK MHD stores magnetic fields on faces. For a cell (i, j, k) , the finite-volume magnetic divergence is

$$(\nabla \cdot \mathbf{B})_{i,j,k} = \frac{B_{i+1/2,j,k}^1 - B_{i-1/2,j,k}^1}{\Delta x_1} + \frac{B_{i,j+1/2,k}^2 - B_{i,j-1/2,k}^2}{\Delta x_2} + \frac{B_{i,j,k+1/2}^3 - B_{i,j,k-1/2}^3}{\Delta x_3}. \quad (1)$$

On a uniform mesh, constrained transport preserves this to roundoff because the same edge-centered electromotive forces update the two cells sharing a face. On AMR, this remains true only if coarse/fine face values are restricted, prolonged, exchanged, and owned consistently. If an active-domain face is written twice, or if internal fine faces are computed from exterior faces that later change, then the local discrete divergence constraint can be broken even though the individual interpolation formulas are mathematically correct.

3 Pre-PR 732 State

Before the PR 732 boundary-ownership work, the face-centered AMR boundary path had no sufficiently general active-face ownership rule. A coarse-neighbor `ProlongateFC` path could write faces that were already owned by a same-level or finer neighbor at the same physical location. This was most visible around coarse/fine edges and corners, where a single prolongation buffer range can include a mixture of true ghost faces and active-domain faces. In 1D this is hard to expose; in 2D and 3D, repeated moving refinement patterns expose it quickly.

The original bug was therefore not “MPI-IO-like nondeterminism” or a failure of CT itself. It was an AMR ownership bug: different boundary paths disagreed about which path owned the final value of a face-centered magnetic field.

The archival pre-732 comparison in Fig. 1 shows the size of the error on the original moving-AMR diagnostic. The 2D and 3D failures were at the 10^{-3} level in normalized maximum divergence,

while the current code is at roundoff. The plotted current 3D value uses the common two-physical AMR-level depth so that the comparison is against the same style of test that originally exposed the bug.

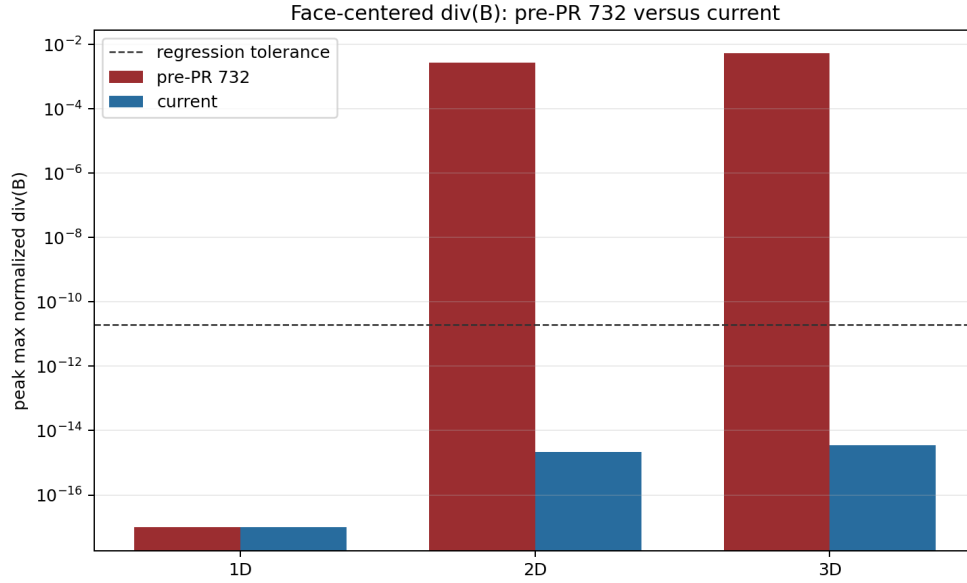


Figure 1: Archival pre-PR 732 moving-AMR result compared with the current code. The current result is many orders of magnitude below the regression tolerance.

4 Why PR 732 Was Necessary But Not Sufficient

PR 732 fixed an important part of the problem: coarse-neighbor boundary prolongation should not overwrite faces owned by same-level or finer neighbors. That change was necessary. However, deeper 3D tests still failed. The failure pattern was the clue: one or two physical AMR levels were clean, while three or more levels produced large $\nabla \cdot \mathbf{B}$ after rebuilds. That behavior pointed away from a simple analytic prolongation error and toward an ordering/ownership error that only appears when refinement and derefinement interact across multiple levels.

The critical remaining issue was stale restricted face data during derefinement. In the mesh-rebuild routine, AMR send buffers are prepared before same-rank derefine copies are performed. Before the current fix, the MHD derefine path could pack or copy `coarse_b0` that had not been freshly restricted from the current `b0`. With shallow AMR this often escapes detection. With deep AMR, stale coarse magnetic faces can seed a new block or a neighbor boundary with a value that is not the restriction of the current fine state. Subsequent CT-consistent operations then preserve the wrong local face balance.

5 Current Code State

The current implementation has four coordinated pieces.

5.1 Refresh Coarse Face Fields Before AMR Packing

After the final `refine_flag` values are known and synchronized, but before `InitRecvAMR()` and `PackAndSendAMR()`, the code now refreshes the MHD coarse face-centered storage when derefine-ment is present:

```
if ((ndel > 0) && (pmhd != nullptr)) {  
    RestrictFC(pmhd->b0, pmhd->coarse_b0);  
}
```

This is the most important correctness fix for the deep-AMR failures. It makes the derefine send/copy source data match the current fine face-centered fields.

5.2 Exact Active-Face Ownership During Boundary Prolongation

`src/bvals/prolongation.cpp` now routes face-centered boundary prolongation writes through `CanProlongateFCFace()` and `StoreProlongatedFCFace()`. The predicate allows ghost-face writes, but it allows active-domain writes only when the face is the normal interface face owned by the coarse-neighbor prolongation path and no same-level or finer neighbor owns that face. Active interior faces are not written by the boundary prolongation path.

This replaces broad range clipping with an exact per-face decision. That matters at edges and corners because a rectangular buffer range can contain both faces that the coarse neighbor should supply and faces that must be left to a same-level/finer neighbor or to the local active state.

5.3 Protect Active Faces During Same/Finer Receives

`MeshBoundaryValuesFC::RecvAndUnpackFC()` now skips same-level and finer receive-buffer writes that target active-domain faces. Direct receive unpacking is still used for ghost data, but it no longer overwrites active magnetic faces through overlapping edge/corner communication paths.

Debug prints for blocked writes are available behind `ATHENAK_DEBUG_FC_AMR_OWNERSHIP`. Production builds pay no always-on provenance cost.

5.4 Post-Boundary AMR Face Repair

`RepairAMRFC()` is called after boundary values and primitives are initialized. It recomputes internal face-centered fields for AMR-affected `MeshBlocks` after exterior faces have been finalized by boundary exchange and prolongation. In multidimensions it reuses the same Toth–Roe internal-face prolongation operator; in 1D it keeps the existing midpoint face average. It does not alter exterior block faces.

The repair mask is broader than “newly refined only.” It marks blocks whose own old source changed level and local blocks adjacent to level-changed neighbors. This is deliberately conservative: the extra work happens only immediately after AMR topology changes, and the affected blocks are precisely where exterior face values may have changed during the rebuild.

After repair, MHD primitives are recomputed from the repaired face fields. This keeps total energy, magnetic pressure, and characteristic speeds consistent with the final magnetic state. No divergence cleaning, projection, or user input control is introduced.

6 Regression and Stress Tests

The validation uses a `divb.amr` problem generator that initializes the field from a discrete vector potential, so the initial face-centered field is divergence-free to roundoff. The refinement pattern is intentionally artificial and moving. It is designed to repeatedly create coarse/fine faces, edges, and corners rather than to follow a passive physical threshold.

The strict pass criteria are

$$\max(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) < 2 \times 10^{-11}, \quad (2)$$

$$L_1(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) < 2 \times 10^{-12}, \quad (3)$$

$$L_2(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) < 5 \times 10^{-12}. \quad (4)$$

6.1 Static Moving-AMR Level Sweep

The deterministic 3D moving-AMR sweep now tests one through five physical AMR levels. The current code passes all depths at roundoff. This is the primary gate because the earlier one- or two-level tests were not discerning enough.

Table 1: Current 3D static moving-AMR level sweep, run with 8 MPI ranks.

Levels	Rows	Peak max	Peak L_1	Peak L_2	Max cells
1	9	3.203e-15	4.812e-17	6.278e-17	115200
2	9	3.576e-15	4.946e-17	6.493e-17	491520
3	9	3.546e-15	4.930e-17	6.483e-17	1839104
4	9	3.898e-15	5.037e-17	6.649e-17	6046720
5	9	3.641e-15	5.046e-17	6.663e-17	15784448

6.2 Driven Turbulence Stress Test

The driven case uses the same 3D AMR diagnostic with MHD turbulent driving:

- 32^3 root grid with 8^3 MeshBlocks;
- three physical refinement levels, giving an effective finest resolution of 256^3 ;
- periodic boundaries, ideal MHD, PLM reconstruction, HLLD;
- `dedt = 75`, `tcorr = 0.002`, `nlow = 1`, `nhigh = 3`;
- completed retained-output Frontier run to $t = 1.600e - 01$.

The completed two-eddy-turnover diagnostic used the diagnostic input under `inputs/tests` with basename `DivBAMR3DTurbE75L3_2Teddy`. It was run on one Frontier node with 8 MPI ranks and one GPU per rank. The input wrote history every 10 cycles, binary slices every 0.02 code time, and full-cube binary diagnostics every 0.08 code time. The run reached $t = 1.600e - 01$ at cycle 1261.

The peak normalized divergences over the completed driven run were

$$\max(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 7.597e - 15, \quad (5)$$

$$L_1(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.158e - 16, \quad (6)$$

$$L_2(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.561e - 16. \quad (7)$$

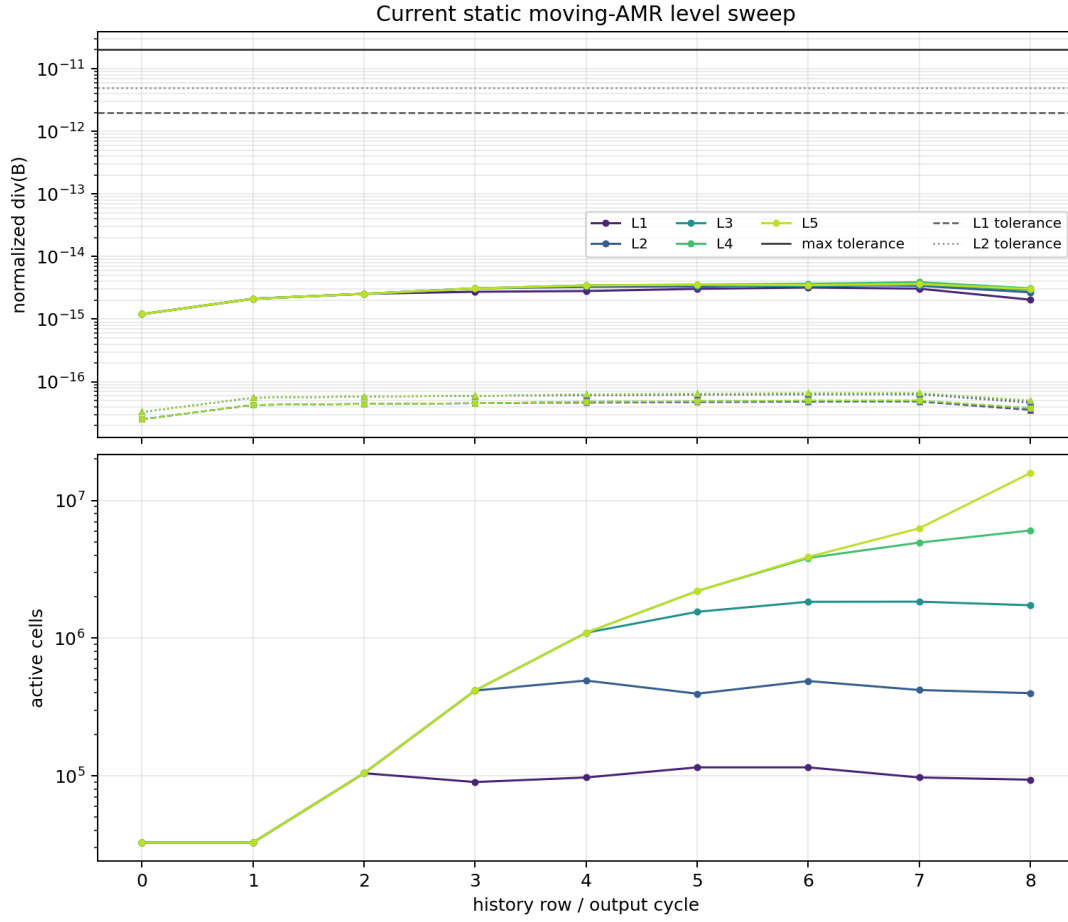


Figure 2: Current static moving-AMR level sweep. All depths remain many orders of magnitude below the strict normalized-divergence tolerances.

The final values were

$$\max(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 6.531e - 15, \quad (8)$$

$$L_1(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.092e - 16, \quad (9)$$

$$L_2(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.469e - 16. \quad (10)$$

The maximum active-cell count was 2097152, and the final active-cell count was 1713664.

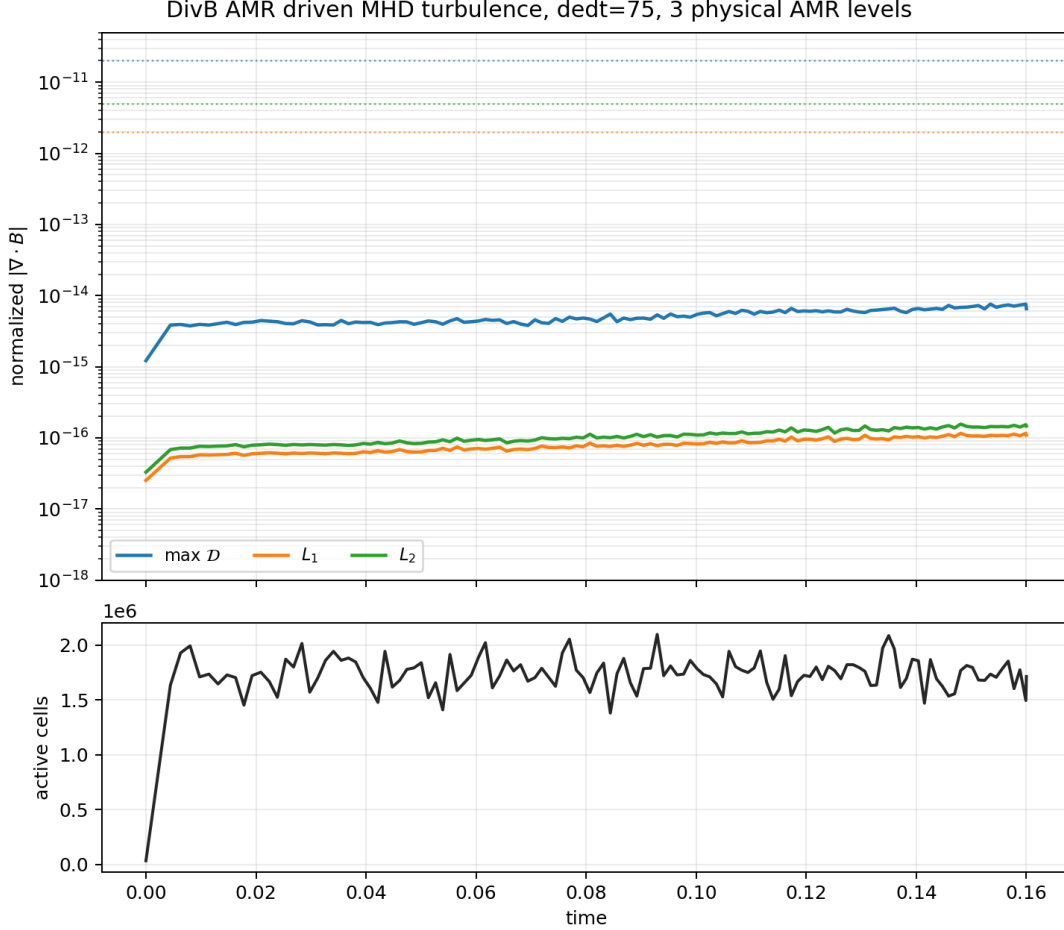


Figure 3: Driven turbulence AMR history for $\text{dedt} = 75$ and three physical refinement levels through $t = 1.600e - 01$. The AMR hierarchy changes aggressively and the divergence diagnostics remain at roundoff throughout the two-eddy-turnover run.

The final full-cube visualization distribution is also clean. On the uniformized final binary cube, the 50th, 95th, and 99th percentiles of $|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}$ are $2.313e - 17$, $8.708e - 17$, and $1.306e - 16$. The maximum in the uniformized visualization cube is $8.055e - 16$. The history file remains the authoritative finite-volume diagnostic because it is accumulated directly over the AMR mesh.

6.3 Compact Moving-Box Four-Level Stress Test

The compact follow-up uses a separate built-in problem generator, `divb_amr_compact`. It keeps the same discrete-vector-potential magnetic initialization and turbulence driving as the three-box driven test, but uses one moving periodic refinement box. The diagnostic run used a 128^3 root grid,

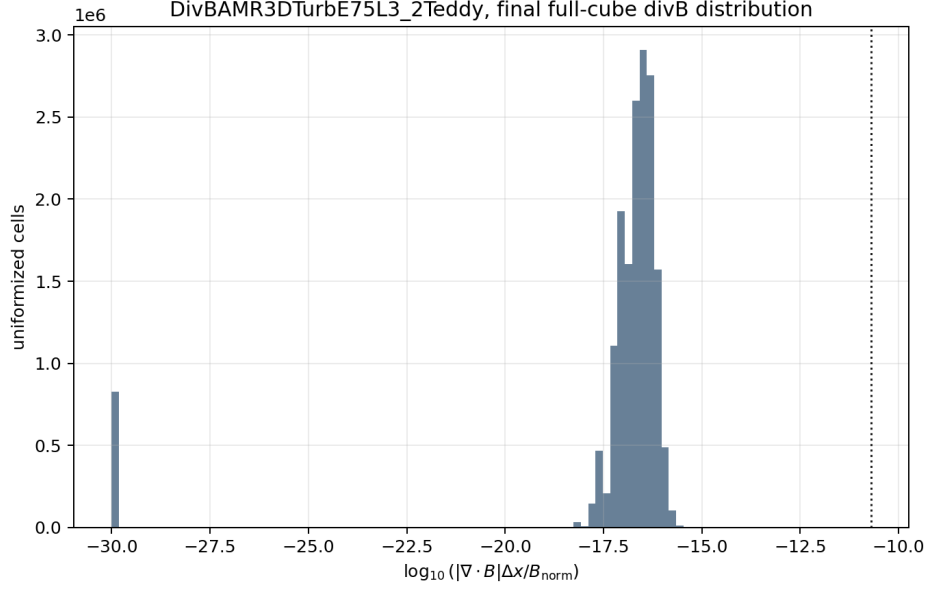


Figure 4: Final full-cube normalized-divergence distribution from the completed driven turbulence diagnostic output.

16^3 MeshBlocks, four physical refinement levels, and `dedt` = 75. The finest effective resolution is 2048^3 , but the output analysis keeps full-cube divergence distributions on native AMR blocks rather than uniformizing to the finest grid.

The completed run used basename `DivBAMRCompact4L128E75_2Teddy` on four Frontier nodes with 32 MPI ranks and one GPU per rank. It reached $t = 1.600e - 01$ at cycle 8115. The peak normalized divergences were

$$\max(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 3.556e - 13, \quad (11)$$

$$L_1(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.819e - 15, \quad (12)$$

$$L_2(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 2.622e - 15. \quad (13)$$

The final values were

$$\max(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 3.553e - 13, \quad (14)$$

$$L_1(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 1.633e - 15, \quad (15)$$

$$L_2(|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}) = 2.431e - 15. \quad (16)$$

These are well below the strict divergence thresholds.

The AMR-occupancy result also satisfies the compact-test intent. The final level-zero volume fraction was $8.750e - 01$, comfortably above the relaxed requirement that more than half of the domain remain at level zero. The final finest-level volume fraction was $2.441e - 04$, inside the requested 10^{-4} – 10^{-3} band, and the peak active-cell count was 11186176, below the 1.2×10^7 cap.

On the final native-AMR full output, the 50th, 95th, and 99th percentiles of $|\nabla \cdot \mathbf{B}| \Delta x / B_{\text{norm}}$ were $3.483e - 16$, $2.395e - 15$, and $5.355e - 15$. The maximum in the native AMR visualization output was $3.239e - 14$.

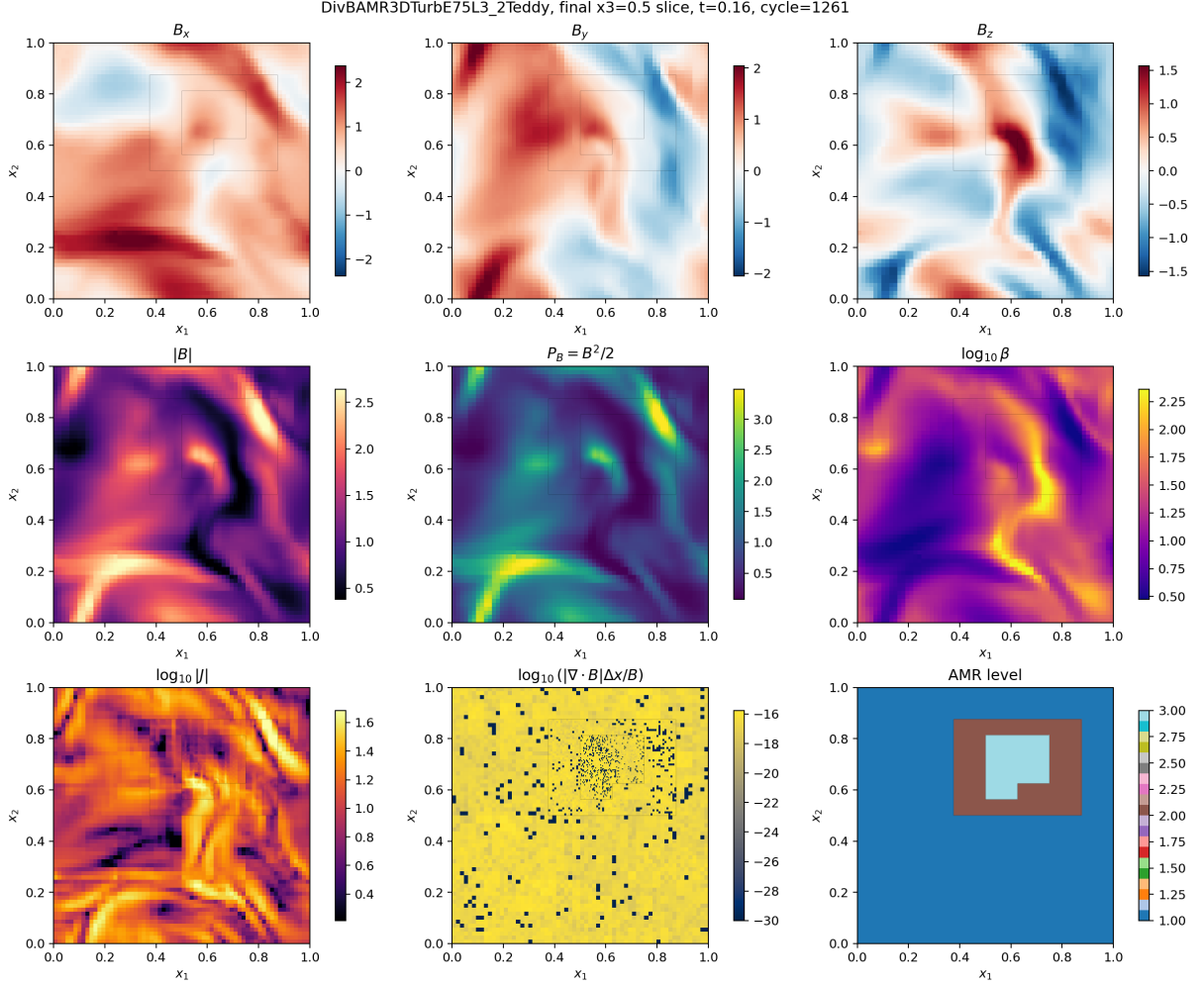


Figure 5: Final $x_3 = 0.5$ slice diagnostics from the driven turbulence test. Magnetic-field structure, current density, and AMR level boundaries are visible, while normalized $\nabla \cdot \mathbf{B}$ remains at roundoff.

Table 2: Final and peak level occupancy for the compact four-level run.

Level	Final volume fraction	Peak volume fraction	Final cells	Peak cells
0	8.750e-01	1.000e+00	1835008	2097152
1	1.094e-01	1.094e-01	1835008	1835008
2	1.367e-02	1.367e-02	1835008	1835008
3	1.709e-03	1.709e-03	1835008	1835008
4	2.441e-04	4.768e-04	2097152	4096000

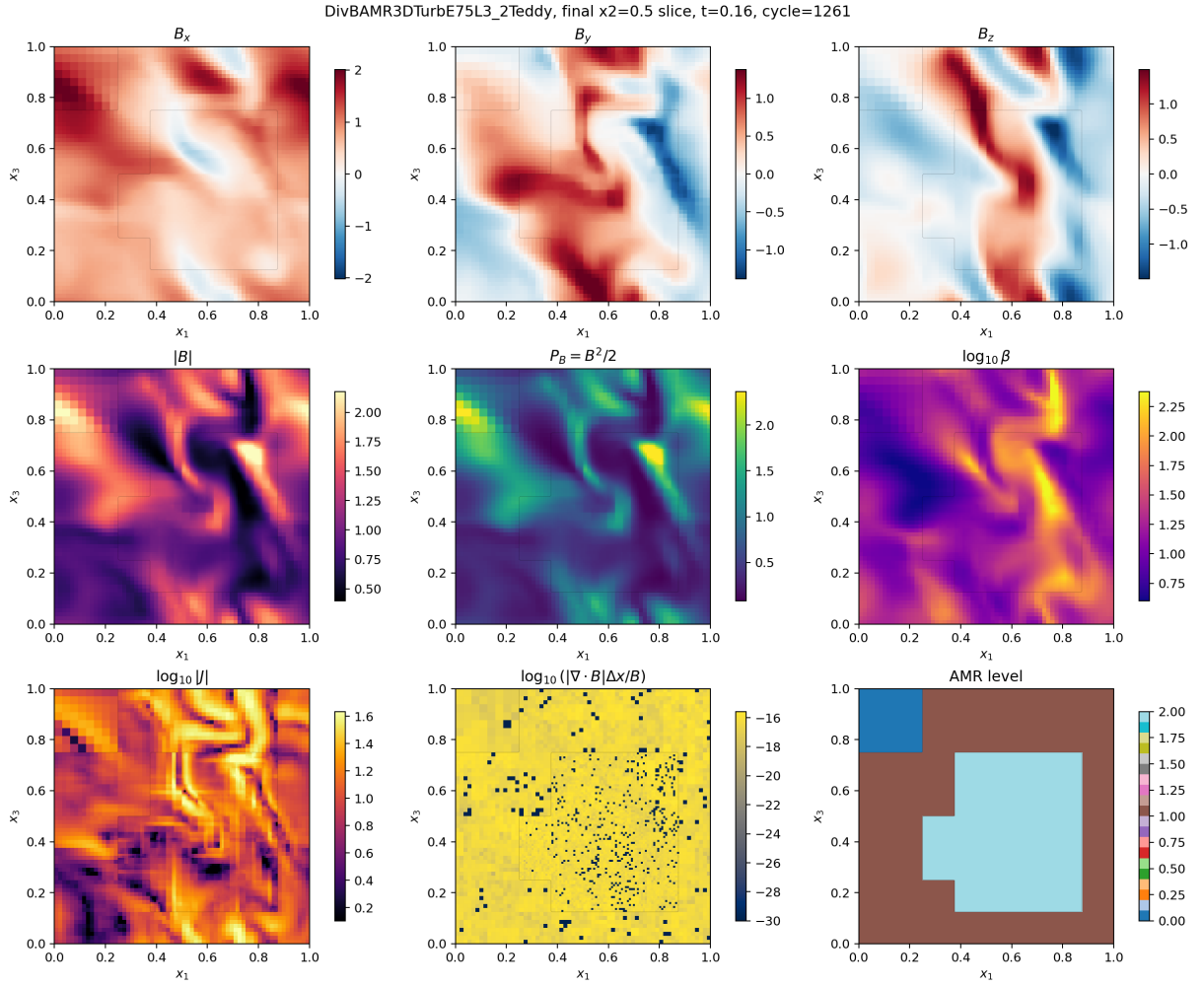


Figure 6: Final $x_2 = 0.5$ slice diagnostics from the same driven turbulence test.

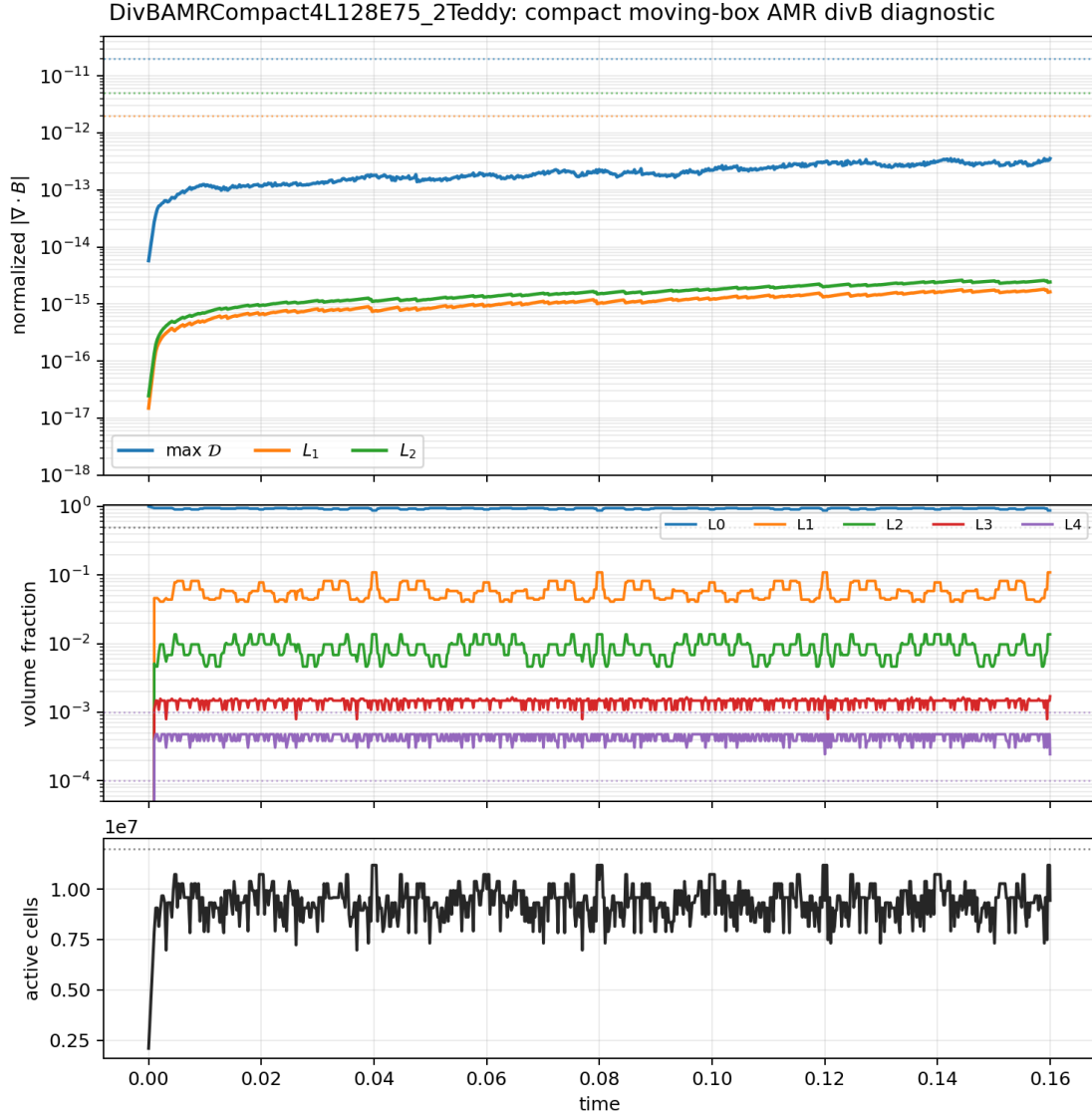


Figure 7: Compact moving-box AMR history. The divergence diagnostics remain far below threshold, while the level-zero volume fraction remains above the compact-test occupancy target.

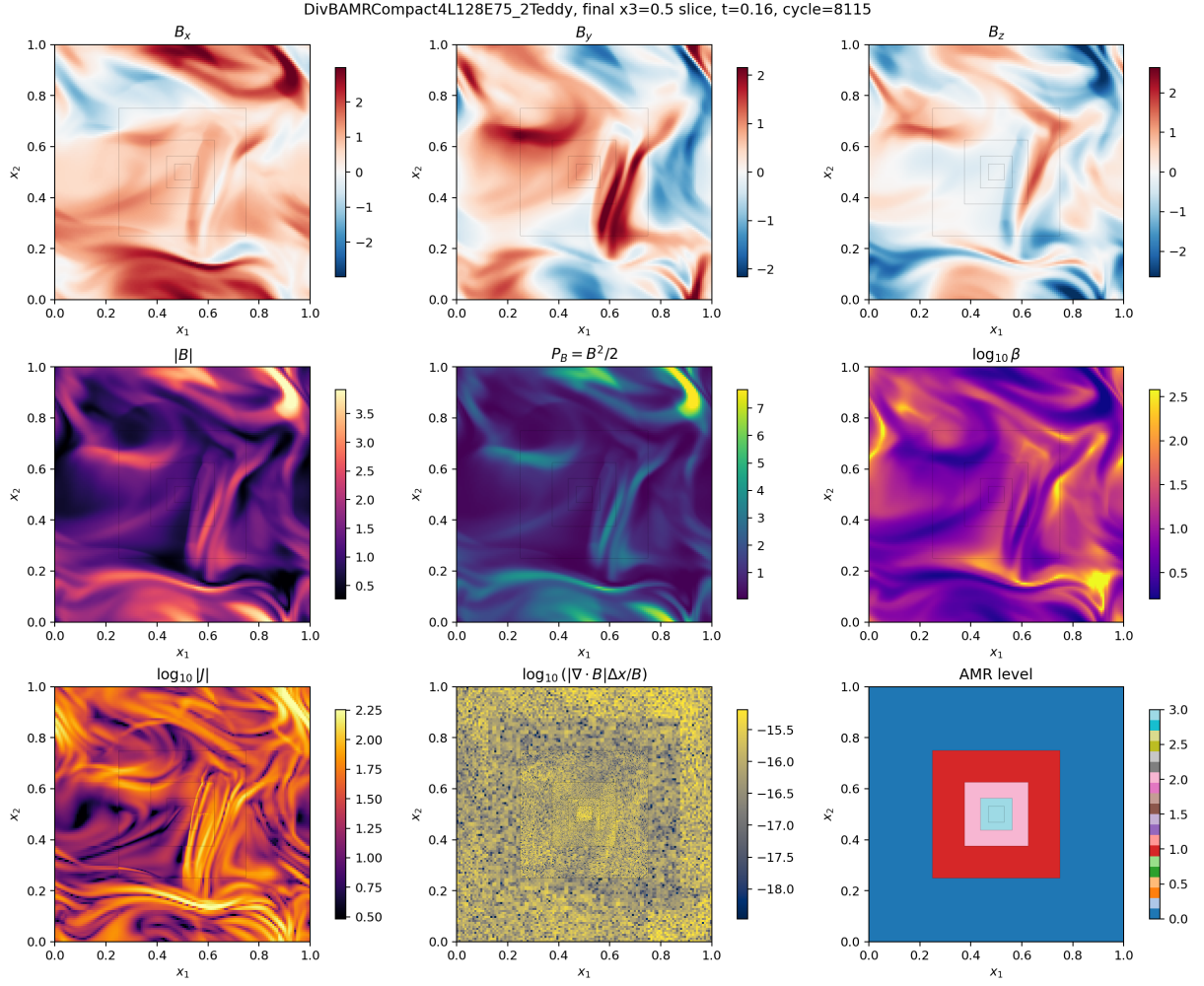


Figure 8: Final $x_3 = 0.5$ slice diagnostics from the compact four-level run. The moving box returns to the domain center at the final time, so this slice cuts through the refined region.

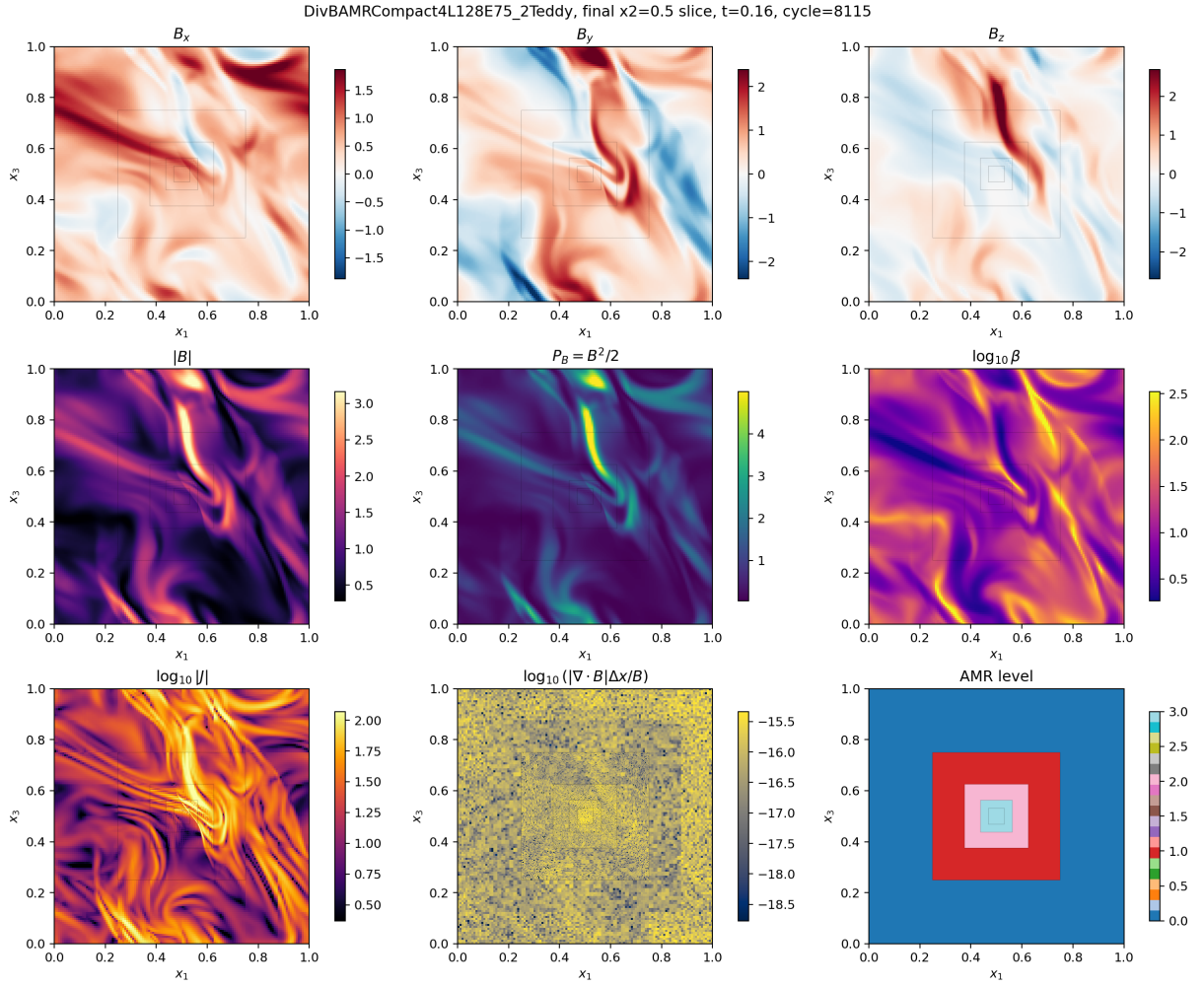


Figure 9: Final $x_2 = 0.5$ slice diagnostics from the compact four-level run.

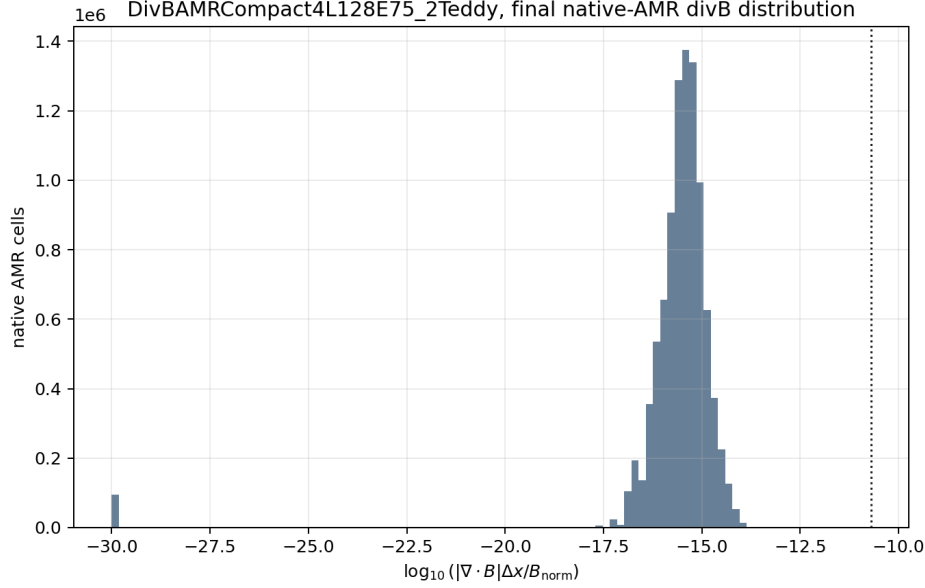


Figure 10: Final native-AMR normalized-divergence distribution for the compact four-level run.

7 Runtime and Speed

The current fixes are deliberately scoped to AMR and boundary work:

- `RestrictFC()` is called before AMR packing only when derefinement occurs and MHD is active.
- The exact face-ownership predicate is evaluated only in face-centered boundary prolongation and only on the coarse-neighbor path.
- `RepairAMRFC()` runs only after AMR topology changes and only on blocks marked by the AMR repair mask.
- `ATHENAK_DEBUG_FC_AMR_OWNERSHIP` diagnostics are off unless the code is built with that macro.

The completed `dedt = 75` diagnostic run used Slurm job 4468997 on one Frontier node. Slurm reported 00:12:57 elapsed wall time. AthenaK reported 4,277,288 MeshBlock-cycles, 762.12 seconds of CPU time, and 2.87×10^6 zone-cycles per CPU-second. The maximum active-cell count was 2097152, with 1713664 active cells at the final history row.

I/O was not limiting. The final output flush reported 3.106 seconds total, with 0.375 seconds spent loading output buffers and 2.809 seconds spent writing. The retained run directory was about 141 MB. History writes were millisecond-scale, slice writes were subsecond, and the full-cube binary writes were also subsecond per field. Memory was not limiting: Slurm allocated 500 GB on the node and reported no out-of-memory condition; the AthenaK log also completed without memory warnings.

The compact four-level run used Slurm job 4469201. Slurm reported 00:08:11 elapsed wall time, and AthenaK reported 18580767 MeshBlock-cycles and $1.614e + 08$ zone-cycles per CPU-second. The final output flush took 3.348 seconds, with 0.105 seconds spent loading output buffers and 3.292 seconds spent writing, so I/O was not limiting. The log and Slurm state show no out-of-memory or abnormal-exit condition; memory did not appear to be a limiting factor for this completed run.

I did not run a same-physics pre-fix versus post-fix performance A/B benchmark in this pass, because the pre-fix code fails the correctness test at three or more physical AMR levels. From the code path and the local timings, the new correctness work does not appear to be a dominant runtime cost. If exact overhead matters for production performance estimates, the right follow-up is a two-binary benchmark with the same problem, same AMR rebuild schedule, and the debug macro disabled in both builds.

8 Practical Checklist for Future Changes

Future MHD AMR changes should preserve the following invariants:

1. A face-centered active-domain face has one owner during an AMR rebuild.
2. Same-level and finer receive/prolongation paths may fill ghost data but must not overwrite active faces.
3. Coarse-face storage used for derefinement must be freshly restricted before AMR send buffers are packed.
4. Toth-Roe internal fine faces must be computed after exterior faces have their final AMR-boundary values.
5. Primitive recovery must run after any face-centered repair that can change magnetic energy or characteristic speeds.

The current deterministic regression should keep the 3D physical level sweep through at least five levels. One or two physical levels are not a sufficient proof of correctness for this bug class.